

1. What is Bash scripting, and why is it useful in Linux administration and DevOps?

Bash scripting is writing a series of Linux commands in a script file so they can run automatically. It is useful because it saves time, reduces manual work, and helps automate system administration and DevOps tasks.

2. What are the basic steps to launch an EC2 instance in AWS?

1. Log in to the AWS Console.
2. Open the EC2 service.
3. Click Launch Instance.
4. Choose an Amazon Machine Image (AMI).
5. Select an instance type.
6. Create or choose a key pair.
7. Configure security groups.
8. Click Launch Instance.

3. How do you connect an AWS EC2 instance to VS Code?

Install the Remote - SSH extension in VS Code, add the EC2 instance details to the SSH config file using its public IP and key pair, then use Remote-SSH: Connect to Host to access the instance.

4. Write a simple shell script to accept user input and display a message.

```
#!/bin/bash
echo "Enter your name:"
read name
echo "Welcome, $name!"
```

5. What kind of tasks can be automated using Bash scripting? Give an example.

Bash scripting can automate tasks like backups, file management, user creation, log monitoring, software installation, and scheduled jobs.

Example: Automatically back up a folder every day using a Bash script and cron.

6. Create a Bash script to display system information such as hostname, memory, and disk usage.

```
#!/bin/bash
```

```
echo "Hostname: $(hostname)"
```

```
echo "Memory Usage:"
```

```
free -h
```

```
echo
```

```
echo "Disk Usage:"
```

```
df -h
```